CO1-AT1

MLA0304-Reinforcement Learning

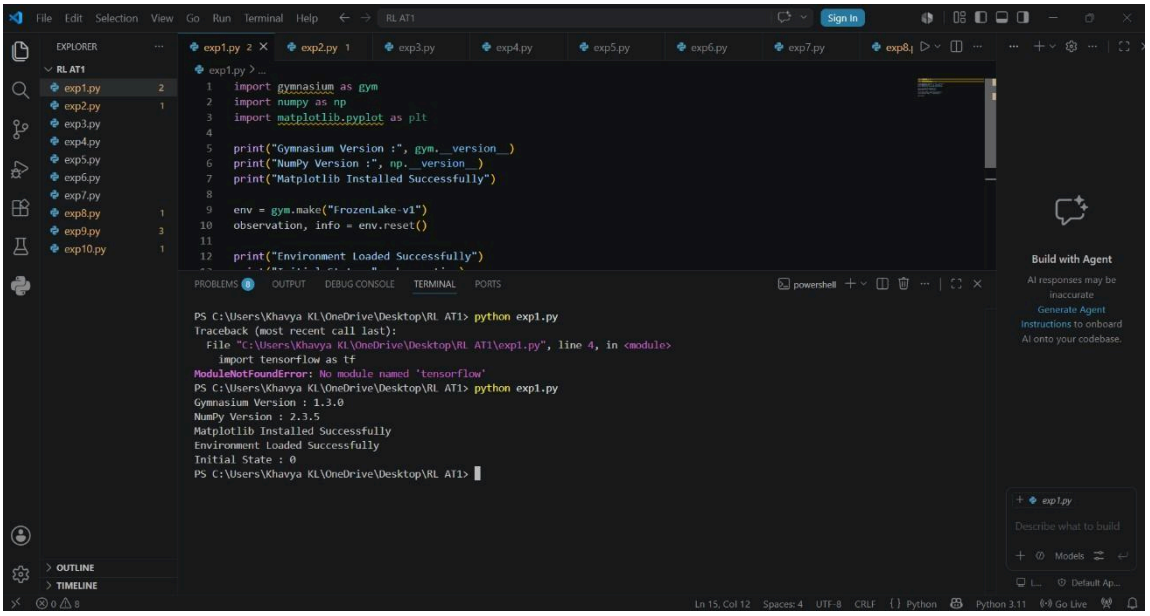
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1. PROGRAM:

```
import gymnasium as gym import numpy as
np import matplotlib.pyplot as plt import
tensorflow as tf import keras print("Gymnasium
Version :", gym.__version__) print("NumPy
Version :", np.__version__) print("TensorFlow
Version :", tf.__version__) print("Keras Version
:", keras.__version__) env =
gym.make("FrozenLake-v1") observation, info
= env.reset() print("Environment Loaded
Successfully") print("Initial State :",
observation) env.close()
```

OUTPUT:



2. PROGRAM:

```
import gymnasium as gym env =
gym.make("FrozenLake-v1", render_mode=None)
```

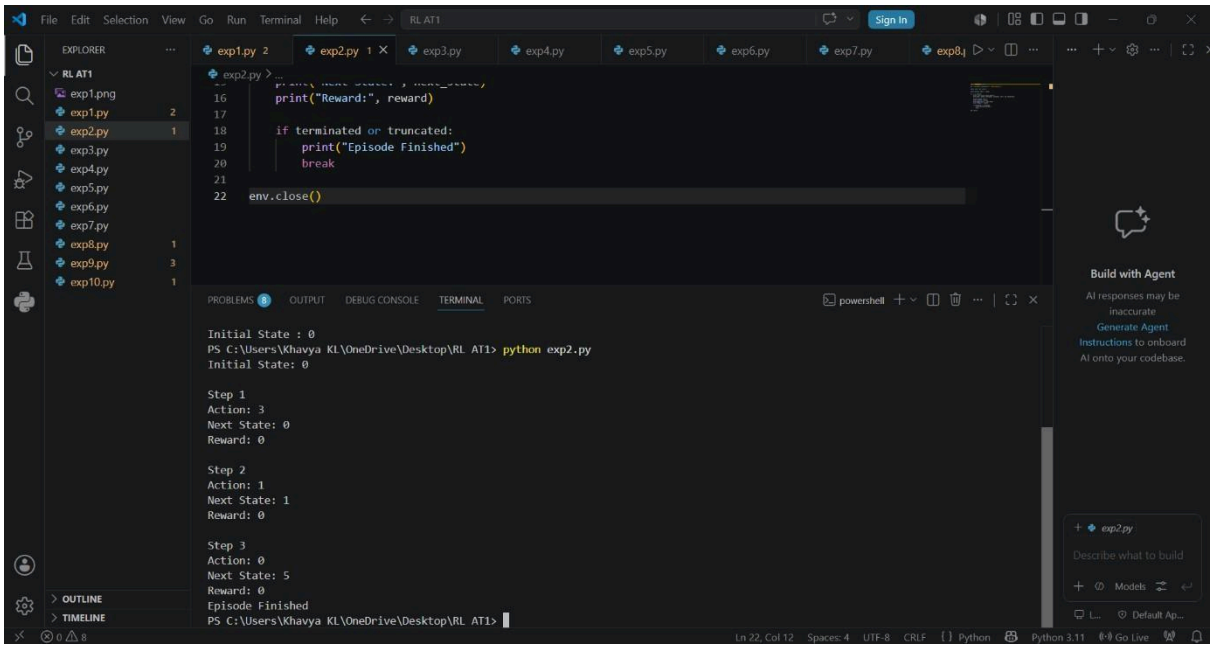
state, info = env.reset()

print("Initial State:", state)

for i in range(5):

action = env.action_space.sample() next_state, reward, terminated, truncated, info = env.step(action) print(f"\nStep {i+1}") print("Action:", action) print("Next State:", next_state) print("Reward:", reward) if terminated or truncated: print("Episode Finished") break env.close()

OUTPUT:



3. PROGRAM:

import random states = ["A", "B", "Goal"] actions = ["Left", "Right"] state = "A" print("Simple Reinforcement Learning Framework\n") while state != "Goal":

action = random.choice(actions)

if state == "A" and action == "Right":

next_state = "B" reward = 5

elif state == "B" and action == "Right":

next_state =

"Goal" reward =

10 else:

next_state = state

reward = -1 print("Current

State :", state) print("Action :",

action) print("Next State :",

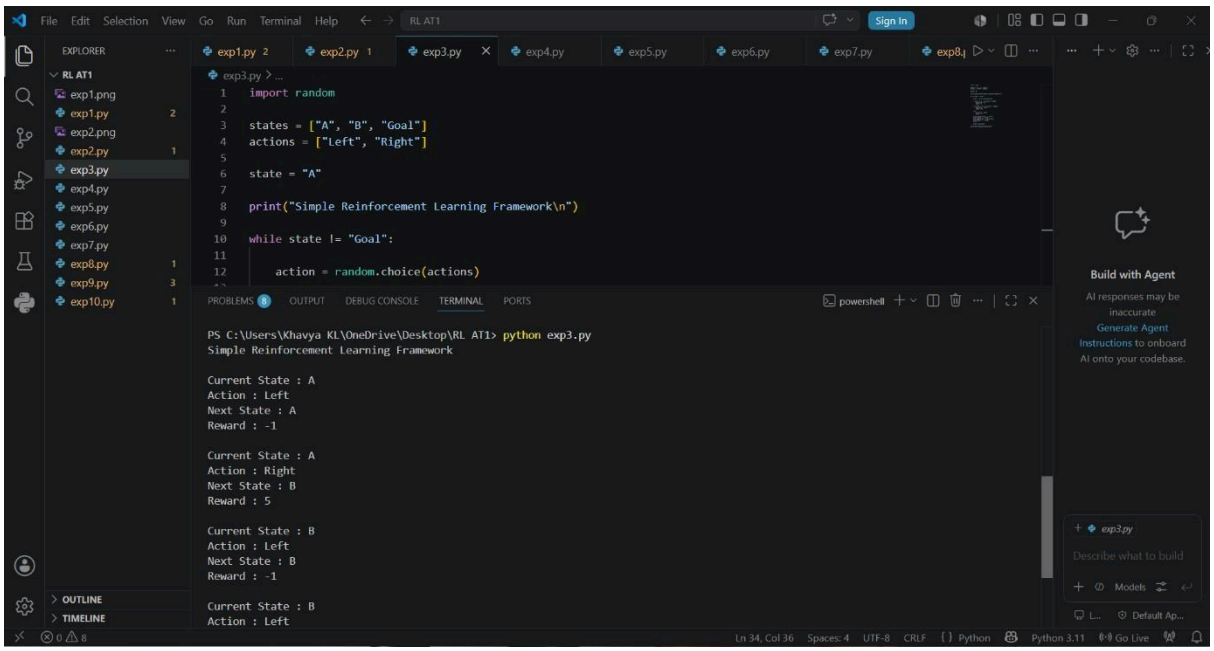
next_state) print("Reward :",

reward)

print() state = next_state

print("Goal Reached

Successfully!") **OUTPUT:**



4. PROGRAM:

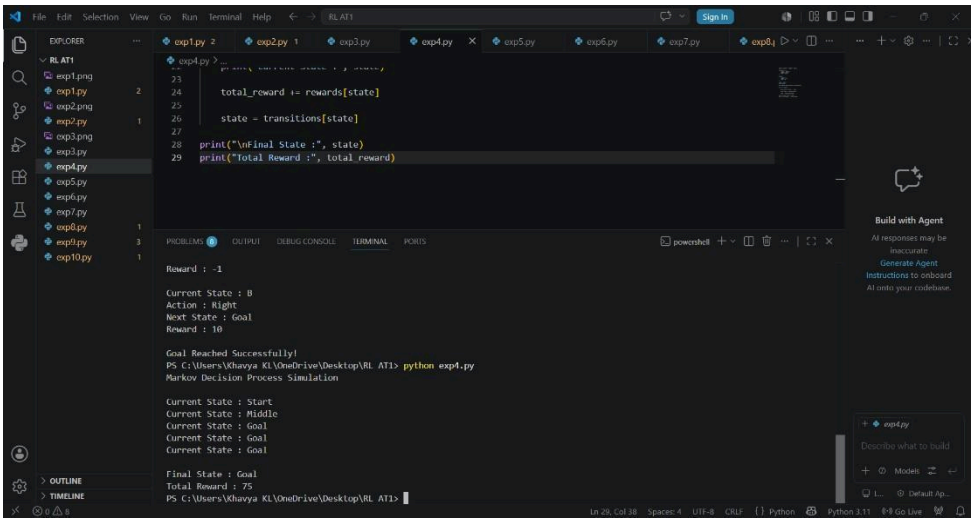
states = ["Start", "Middle",

"Goal"] transitions = {

"Start": "Middle",

```
        "Middle": "Goal",
"Goal": "Goal"} rewards =
{
        "Start": 5,
        "Middle": 10,
        "Goal": 20
    } state = "Start" total_reward = 0
print("Markov Decision Process Simulation\n")
for i in range(5):
        print("Current State :", state)
total_reward += rewards[state]
state = transitions[state]
print("\nFinal State :", state)
print("Total Reward :", total_reward)
```

OUTPUT:

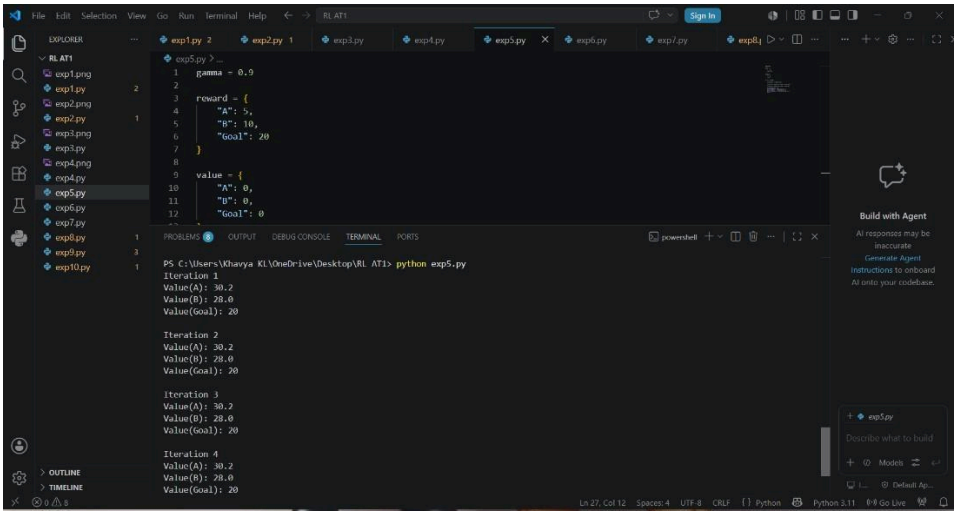


5**. PROGRAM:**

```
gamma = 0.9
reward = {
        "A": 5,
        "B": 10,
        "Goal": 20
    }
value =
{
        "A": 0,
        "B": 0,
        "Goal": 0
    } for i in
range(5):
        value["Goal"] = reward["Goal"]     value["B"] =
reward["B"] + gamma * value["Goal"]     value["A"]
= reward["A"] + gamma * value["B"]
print("Iteration", i+1)     print("Value(A):",
round(value["A"],2))     print("Value(B):",
round(value["B"],2))     print("Value(Goal):",
round(value["Goal"],2))
        print()
```

OUTPUT:

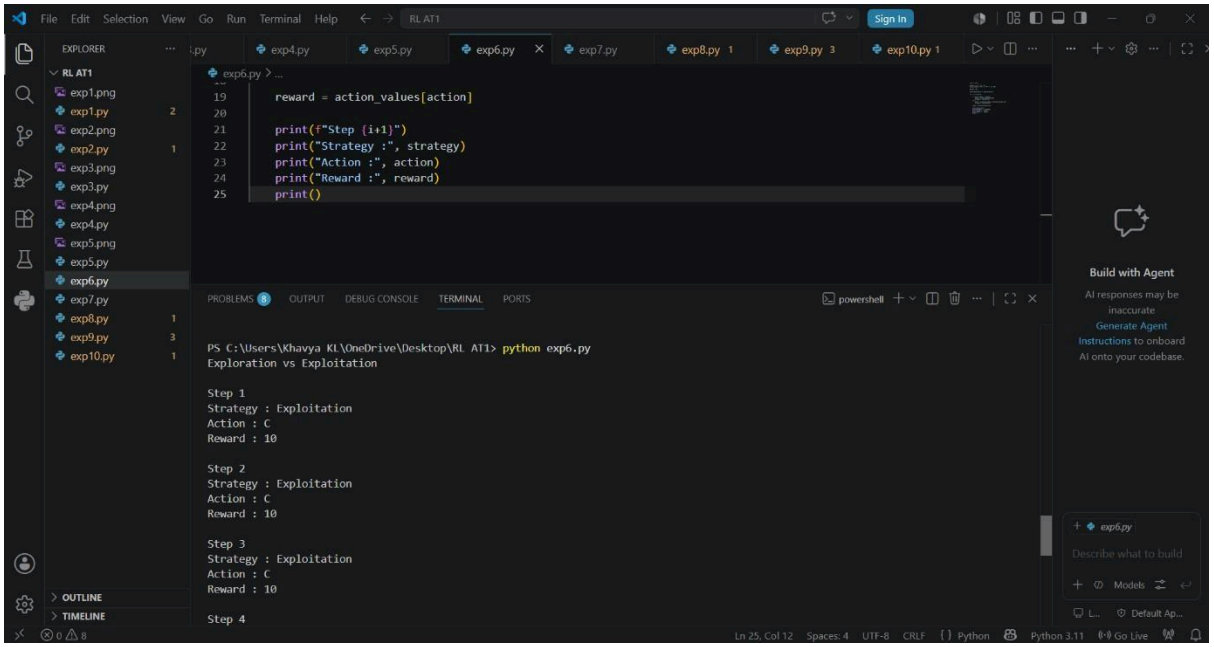
6. **PROGRAM:**



import random actions = ["A", "B", "C"] action_values = {"A": 5, "B": 8, "C": 10} epsilon = 0.2 print("Exploration vs Exploitation\n") for i in range(10): if random.random() < epsilon: action = random.choice(actions) strategy = "Exploration" else: action = max(action_values, key=action_values.get) strategy = "Exploitation" reward = action_values[action] print(f"Step {i+1}") print("Strategy :", strategy) print("Action :", action) print("Reward :", reward) print()

OUTPUT:

7. **PROGRAM:**



import random arms = [2, 5, 8, 10]

print("Multi-Armed Bandit Problem\n")

total_reward = 0 for i in range(10):

arm = random.randint(0, 3)

reward = arms[arm] total_reward

+= reward print("Round", i+1)

print("Selected Arm :", arm+1)

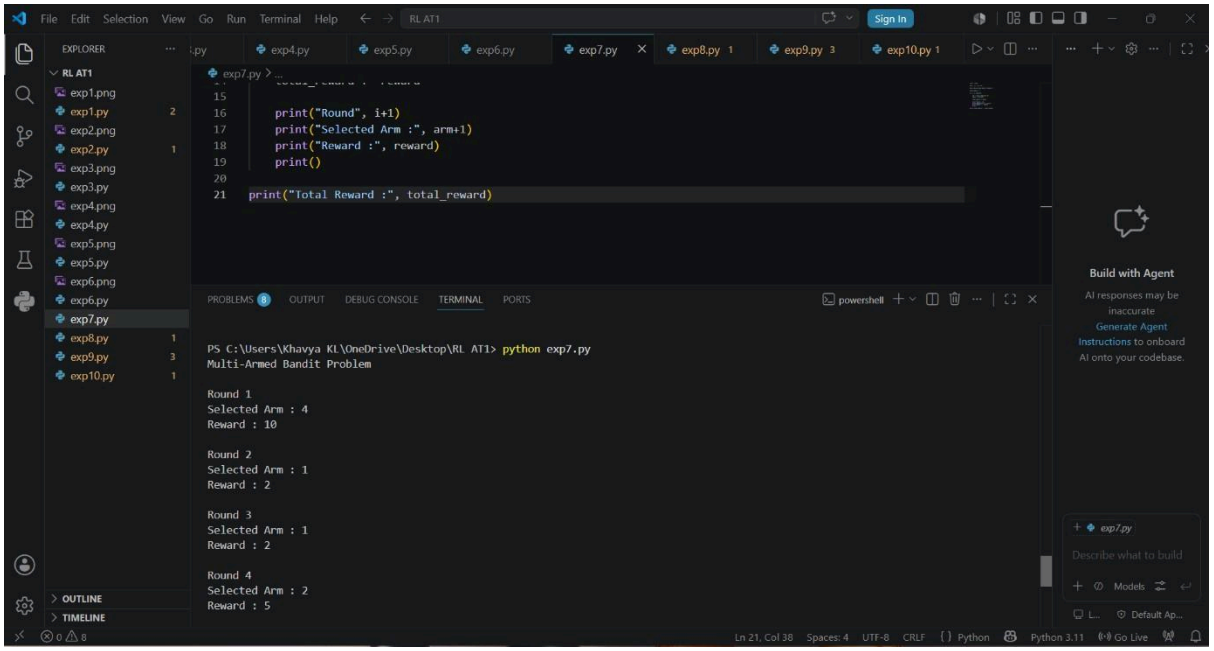
print("Reward :", reward)

print() print("Total Reward :",

total_reward)

OUTPUT:

8. **PROGRAM:**




```
import matplotlib.pyplot as plt

episode = [1,2,3,4,5,6,7,8,9,10]
reward =
[5,8,12,15,20,25,28,30,35,40]

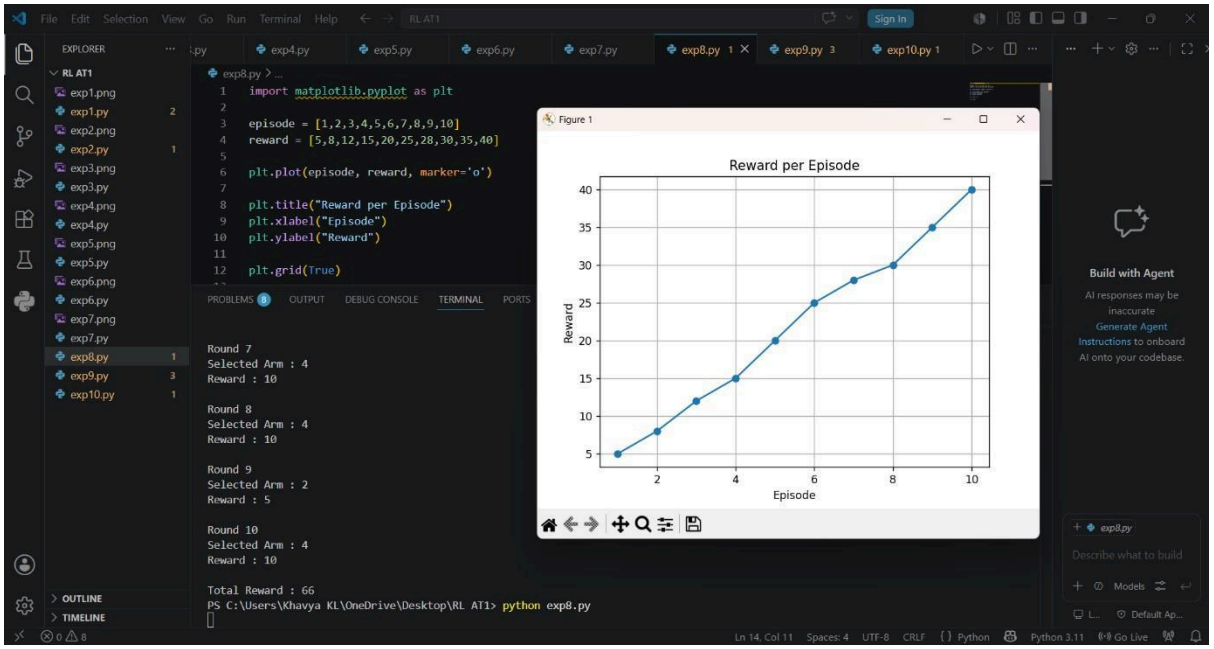
plt.plot(episode, reward, marker='o')

plt.title("Reward per Episode")
plt.xlabel("Episode")
plt.ylabel("Reward")

plt.grid(True)
plt.show()
```

9. **PROGRAM:**

OUTPUT:



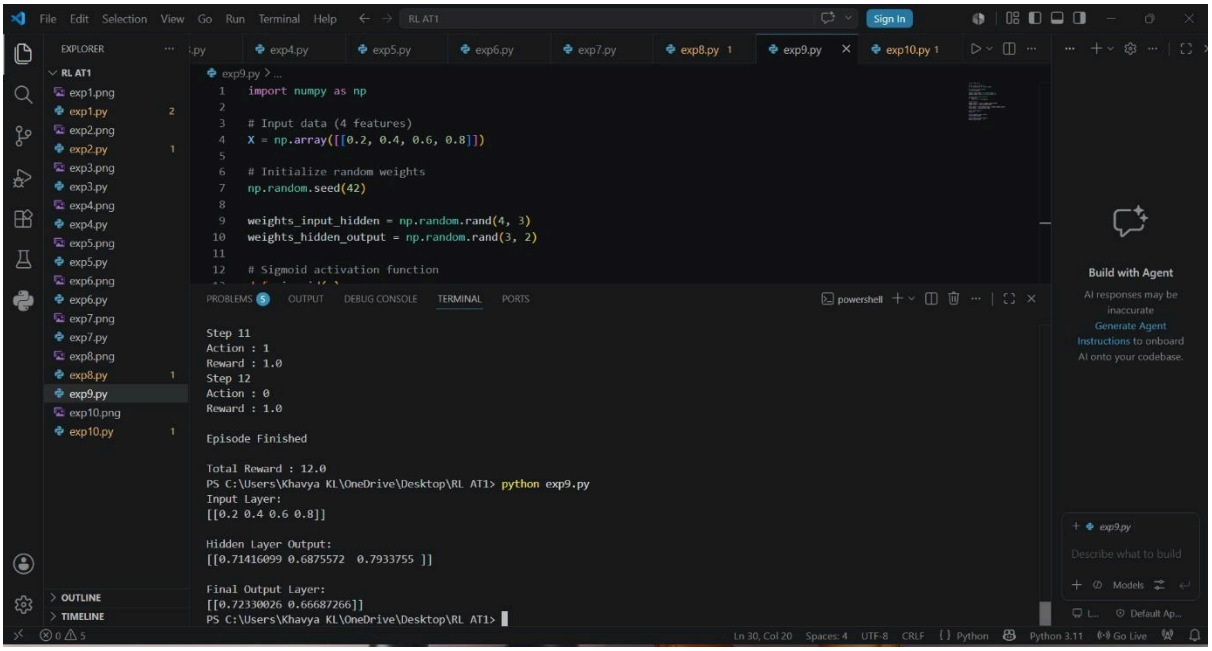
import tensorflow as tf from tensorflow.keras.models import Sequential from tensorflow.keras.layers import Dense model = Sequential() model.add(Dense(16, activation='relu', input_shape=(4,)))

model.add(Dense(16, activation='relu'))

model.add(Dense(2, activation='linear'))

model.summary() **OUTPUT:**

10. **PROGRAM:**



import gymnasium as gym

env =

gym.make("CartPole-v1")

state, info = env.reset()

print("CartPole Simulation\n")

total_reward = 0 for step in

range(20):

action = env.action_space.sample() state, reward,

terminated, truncated, info = env.step(action) total_reward

11**. PROGRAM:**

+= reward print(f"Step {step+1}") print("Action :", action) print("Reward :", reward) if terminated or truncated:

print("\nEpisode Finished")

break print("\nTotal Reward :", total_reward) env.close()

OUTPUT:

